Ben Gu

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA
M.S. in Robotics, Robotics Institute — *GPA: 4.0/4.0* — Aug 2025 – May 2027
B.S. in Mechanical Engineering, College of Engineering — *GPA: 4.0/4.0* — Aug 2022 – May 2025

EXPERIENCE

Physical AI Engineer Intern — May 2026 – Aug 2026
Stealth Robotics Startup, Boston, MA

- Trained a **PPO** posture-control policy to replace a model-based QP/IK controller; deployed on hardware and refined with real-world data, validating on three outdoor terrains unseen in training (grass, gravel, pavement).
- Benchmarked three architectures (**reactive**, **recurrent**, and **teacher-distilled**) with identical reward and observation spaces; showed that direct RL on the recurrent policy reached deployment without two-stage distillation.
- Built the platform end-to-end: mechanical and mechatronic design, a low-level base controller over **CAN**, and a data pipeline to onboard compute.

RESEARCH

Student Researcher, *Robomechanics Lab, CMU (PI: Prof. Aaron Johnson)* — Dec 2023 – Present

FORLAX: First-Order Reinforcement Learning in JAX — Sep 2026 – Present

- Leading the **Unitree G1** humanoid velocity-tracking and **LEAP Hand** in-hand reorientation tasks: converting non-differentiable reward and termination terms into **smooth surrogates**, tuning **first-order** training, and benchmarking against zeroth-order baselines (**PPO**) on sample efficiency and wall-clock time.

Self-Stabilizing Penguin-Inspired Bipedal Robot for Low-Friction Slopes — Aug 2025 – Present

- Designed (**SolidWorks**) and built a 5-DOF penguin-inspired biped with a crank-slider leg extension and adjustable center-of-mass height; modeled its closed-chain leg mechanism in **MuJoCo** and **NVIDIA Isaac Lab**.
- Swept gait parameters across four friction coefficients in MuJoCo and validated on hardware with **motion capture**: penguin-style torso-over-stance-leg motion with a high COM was 1.8× faster with 44% lower cost of transport than an upright human-like gait at $\mu = 0.12$, reversing at $\mu = 0.45$.
- Extended the study to closed-loop control: trained **CPG-RL** and **imitation RL** policies in Isaac Lab that climb 6° slopes at $\mu = 0.1$ in simulation.
- **Publications (Co-First Author):** "When to Waddle: A Comparative Study of Bipedal Torso-Stabilization on Low-Friction Surfaces," arXiv:2609.21185, under review at **ICRA 2027**; workshop papers at **IROS 2026** and **ICRA 2026**; live hardware demos at **NFL Draft Tech Day**, **Robotics & AI Discovery Day**, and **NSF Foundational Research in Robotics (FRR)**, all in Pittsburgh, 2026.

Scaling Laws for Minimally Actuated Bipeds — Dec 2023 – Sep 2026

- Engineered a **PID attitude controller** for the first steerable single-motor biped; analyzed its 2D contact dynamics and steering manifolds in **MATLAB** for limit-cycle stability and gait efficiency.
- Designed and built hardware and **Drake** models of both Mugatu and Zippy (2.5 cm leg, the smallest power-autonomous biped); swept foot shape and scale across 0.02–1.2 m leg lengths, confirming speed $\propto L^{0.5}$ and hip torque $\propto mL$.
- **Publication:** "Allometric Scaling Laws for Bipedal Robots," **Co-Author**, arXiv:2603.22560, under review at **ICRA 2027**; presented at **Gordon Research Conference on Robotics 2026**, **NCUR 2025**, and **CMU MOTM 2025**.

PROJECTS

Hybrid Locomotion for Unitree Go2-W in Agricultural Environments — Jan 2026 – Sep 2026
Course Project, Robotics & AI for Agriculture (Instructors: Prof. George Kantor & Dr. Francisco Yandun)

- Trained a **PPO** policy in **NVIDIA Isaac Lab** for adaptive wheel-leg switching on the Go2-W, enabling robust navigation in procedurally generated off-road terrain with multi-crop row constraints.

Sim-to-Real RL: Autonomous Quadrotor Landing — Aug 2025 – Jan 2026
Course Project, Robot Learning (Instructor: Prof. Guanya Shi)

- Trained a **PPO** policy using a curriculum with exponential reward scaling for precision landing on moving targets (<2 cm error); mitigated the sim-to-real gap, achieving a **96%** success rate on Crazyflie 2.1+ hardware.

SKILLS

- **Languages:** Python, C++, C, MATLAB, Julia **Embedded:** Arduino, Raspberry Pi, Dynamixel SDK, CAN, IMU
- **Tools:** MuJoCo, Isaac Lab, Drake, PyBullet, Dojo, MJPC (MPC), PyTorch, JAX, ROS 2, Quad-SDK
- **Methods:** Reinforcement Learning (PPO, SAC, DQN), Differentiable Simulation, MPC (MPPI), LQR, CPG
- **Mechanical:** SolidWorks, Onshape, Blender, ANSYS (FEA), CNC (mill, lathe), waterjet, laser cutting, 3D printing